

DataImportFromPCIBex

Write your name here

2026-05-01

1. Load the function to import the dataset

```
read.pcibex <- function(filepath, auto.colnames=TRUE, fun.col=function(col,cols){cols[cols==col]<-paste0(col,"_",col)}, na.rm=TRUE){
  n.cols <- max(count.fields(filepath,sep=","),na.rm=TRUE)
  if (auto.colnames){
    cols <- c()
    con <- file(filepath, "r")
    while ( TRUE ) {
      line <- readLines(con, n = 1, warn=FALSE)
      if ( length(line) == 0 ) {
        break
      }
      m <- regmatches(line,regexec("^# (\\d+)\\. (.+)\\.\\.$",line))[[1]]
      if (length(m) == 3) {
        index <- as.numeric(m[2])
        value <- m[3]
        if (is.function(fun.col)){
          cols <- fun.col(value,cols)
        }
        cols[index] <- value
        if (index == n.cols){
          break
        }
      }
    }
    close(con)
    return(read.csv(filepath, comment.char="#", header=FALSE, col.names=cols))
  }
  else{
    return(read.csv(filepath, comment.char="#", header=FALSE, col.names=seq(1:n.cols)))
  }
}
```

2. Import the data using this function

2.1 Import the data

```
data <- read.pcihex("results_FAKE_FOR_CHC9012.csv")
```

2.2 Have a glimpse at it

```
head(data, 20)
```

```
##      Results.reception.time MD5.hash.of.participant.s.IP.address Controller.name
## 1      1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 2      1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 3      1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 4      1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 5      1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 6      1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 7      1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 8      1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 9      1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 10     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 11     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 12     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 13     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 14     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 15     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 16     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 17     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 18     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 19     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
## 20     1777517421      965c6743ad440dd6eaf441586b84d5c2 PennController
##      Order.number.of.item Inner.element.number      Label
## 1              0              0      Welcome
## 2              0              0      Welcome
## 3              0              0      Welcome
## 4              0              0      Welcome
## 5              1              0      Background
## 6              1              0      Background
## 7              1              0      Background
## 8              1              0      Background
## 9              1              0      Background
## 10             1              0      Background
## 11             2              0      Instructions
## 12             2              0      Instructions
## 13             2              0      Instructions
## 14             2              0      Instructions
## 15             3              0      AnnouncePractice
## 16             3              0      AnnouncePractice
## 17             3              0      AnnouncePractice
## 18             3              0      AnnouncePractice
```

```

## 19          4          0      Practice
## 20          4          0      Practice
##   Latin.Square.Group PennElementType PennElementName Parameter Value
## 1          NULL PennController          0   _Trial_ Start
## 2          NULL PennController          0   _Header_ Start
## 3          NULL PennController          0   _Header_ End
## 4          NULL PennController          0   _Trial_ End
## 5          NULL PennController          1   _Trial_ Start
## 6          NULL PennController          1   _Header_ Start
## 7          NULL PennController          1   _Header_ End
## 8          NULL      TextInput          Age      Final    30
## 9          NULL      DropDown          Degree Selected Ph.D.
## 10         NULL PennController          1   _Trial_ End
## 11         NULL PennController          2   _Trial_ Start
## 12         NULL PennController          2   _Header_ Start
## 13         NULL PennController          2   _Header_ End
## 14         NULL PennController          2   _Trial_ End
## 15         NULL PennController          3   _Trial_ Start
## 16         NULL PennController          3   _Header_ Start
## 17         NULL PennController          3   _Header_ End
## 18         NULL PennController          3   _Trial_ End
## 19         NULL PennController          4   _Trial_ Start
## 20         NULL PennController          4   _Header_ Start
##      EventTime ItemNo Condition Group UserAge UserDegree Comments
## 1  1.777517e+12   NULL          NA
## 2  1.777517e+12   NULL          NA
## 3  1.777517e+12   NULL          NA
## 4  1.777517e+12   NULL          NA
## 5  1.777517e+12   NULL          NA
## 6  1.777517e+12   NULL          NA
## 7  1.777517e+12   NULL          NA
## 8  1.777517e+12   NULL          NA
## 9  1.777517e+12     3          NA
## 10 1.777517e+12   NULL          NA
## 11 1.777517e+12   NULL          NA
## 12 1.777517e+12   NULL          NA
## 13 1.777517e+12   NULL          NA
## 14 1.777517e+12   NULL          NA
## 15 1.777517e+12   NULL          NA
## 16 1.777517e+12   NULL          NA
## 17 1.777517e+12   NULL          NA
## 18 1.777517e+12   NULL          NA
## 19 1.777517e+12   NULL          NA
## 20 1.777517e+12   NULL          NA

```

```
str(data)
```

```

## 'data.frame':   6195 obs. of  18 variables:
## $ Results.reception.time      : int  1777517421 1777517421 1777517421 1777517421 1777517421
## $ MD5.hash.of.participant.s.IP.address: chr  "965c6743ad440dd6eaf441586b84d5c2" "965c6743ad440dd6eaf441586b84d5c2"
## $ Controller.name             : chr  "PennController" "PennController" "PennController" "PennController"
## $ Order.number.of.item        : int  0 0 0 0 1 1 1 1 1 1 1 ...
## $ Inner.element.number        : int  0 0 0 0 0 0 0 0 0 0 0 ...
## $ Label                       : chr  "Welcome" "Welcome" "Welcome" "Welcome" ...

```

```
## $ Latin.Square.Group          : chr "NULL" "NULL" "NULL" "NULL" ...
## $ PennElementType             : chr "PennController" "PennController" "PennController" "PennController" ...
## $ PennElementName             : chr "0" "0" "0" "0" ...
## $ Parameter                   : chr "_Trial_" "_Header_" "_Header_" "_Trial_" ...
## $ Value                       : chr "Start" "Start" "End" "End" ...
## $ EventTime                   : num 1.78e+12 1.78e+12 1.78e+12 1.78e+12 1.78e+12 ...
## $ ItemNo                      : chr "NULL" "NULL" "NULL" "NULL" ...
## $ Condition                   : chr "" "" "" "" ...
## $ Group                       : chr "" "" "" "" ...
## $ UserAge                     : int NA NA NA NA NA NA NA NA NA NA ...
## $ UserDegree                  : chr "" "" "" "" ...
## $ Comments                    : chr "" "" "" "" ...
```

```
summary(data)
```

```
## Results.reception.time MD5.hash.of.participant.s.IP.address Controller.name
## Min. :1.778e+09      Length:6195      Length:6195
## 1st Qu.:1.778e+09    Class :character      Class :character
## Median :1.778e+09    Mode  :character      Mode  :character
## Mean   :1.778e+09
## 3rd Qu.:1.778e+09
## Max.   :1.778e+09
##
## Order.number.of.item Inner.element.number Label
## Min. : 0.00      Min. :0      Length:6195
## 1st Qu.: 9.00    1st Qu.:0      Class :character
## Median :18.00    Median :0      Mode  :character
## Mean   :17.75    Mean :0
## 3rd Qu.:27.00    3rd Qu.:0
## Max.   :35.00    Max. :0
##
## Latin.Square.Group PennElementType PennElementName Parameter
## Length:6195      Length:6195      Length:6195      Length:6195
## Class :character  Class :character  Class :character  Class :character
## Mode :character   Mode :character   Mode :character   Mode :character
##
##
##
## Value           EventTime           ItemNo           Condition
## Length:6195     Min. :1.778e+12  Length:6195     Length:6195
## Class :character 1st Qu.:1.778e+12  Class :character  Class :character
## Mode :character  Median :1.778e+12  Mode :character  Mode :character
## Mean :1.778e+12
## 3rd Qu.:1.778e+12
## Max. :1.778e+12
##
## Group           UserAge           UserDegree        Comments
## Length:6195     Min. :19.00      Length:6195      Length:6195
## Class :character 1st Qu.:27.00    Class :character  Class :character
## Mode :character  Median :33.00    Mode :character  Mode :character
## Mean :37.86
## 3rd Qu.:51.00
## Max. :65.00
```

##

NA's :945